Unit 5 discoveries and equations reference

public-student | All competition levels | Modern notation with historical limits

Black: material response and phase change

Q=mcΔT applies to sensible heating with an appropriate approximately constant c and no phase change. Q is J; m is kg; c is J/(kg·K); ΔT is K or a Celsius difference. Q=mL applies to the stated phase change, with L in J/kg. For 0.10 kg water, c=4200 J/(kg·K) and ΔT=2 K, Q=840 J. For 0.010 kg melting ice with L=334000 J/kg, Q=3340 J. These modern relationships express distinctions associated with Black; they are not a transcription of his notation. Heat capacity C=mc (J/K) differs from conductivity (W/(m·K)).

Kelvin: an absolute temperature framework

T=t+273.15 converts Celsius t to Kelvin T. At 0 °C, T=273.15 K. A 5 °C temperature change is 5 K, but a ratio of Celsius readings is not an absolute-temperature ratio. Kelvin’s 1848 proposal and the present SI definition are different historical stages.

Joule: conservation in common energy units

ΔU=Q−W uses heat into a closed system as positive Q and work by it as positive W; neglect bulk kinetic and potential energy changes. A 100 J work input with Q=0 means W=−100 J and ΔU=+100 J. One J equals one N·m. The first law had multiple contributors; Joule supplied major quantitative experimental evidence.

Carnot: actual performance versus a modern reversible limit

W_net=Q_H−Q_C and η=W_net/Q_H use positive heat magnitudes. The modern ideal bound is η_max=1−T_C/T_H, with Kelvin reservoir temperatures. Between 600 K and 300 K the ceiling is 50%. Carnot’s original published caloric framework did not use the complete modern energy interpretation. The bound is a limit, not a guarantee of actual performance.

Clausius: entropy and direction

For an isolated total, ΔS_total≥0. S has units J/K, not J. A reversible ideal process gives zero total change; an irreversible one gives a positive change. Local decreases are possible. Adding supplied changes −1 and +2 J/K gives a total +1 J/K. No advanced entropy formula is required here.

Maxwell: distributions, not one universal molecular speed

The number in a bin divided by the total gives that bin’s fraction. In a 100-particle teaching table, 25 particles in a bin means 25%. Equal-width bins allow direct height comparisons. A stable distribution does not imply motionless particles, and hotter distributions can overlap cooler ones.

Boltzmann: multiplicity and a temperature-energy link

Count compatible arrangements under stated assumptions. Four independent two-choice labeled tokens give $2^4=16$ assignments. Six are balanced two-and-two, so their combined probability is $6/16=37.5\%$. This is a finite teaching analogy, not a derivation of gas entropy. k_B has units J/K; $k_B T$ has energy units. No logarithmic entropy calculation is assessed.

Planck: discrete energy elements

$\varepsilon=hf$ expresses the conceptual relation between an energy element ε (J), frequency f (Hz) and Planck's constant h (J·s). Doubling f doubles ε at fixed h. This does not determine total lamp power. The oscillator-energy contribution is distinguished from later photon theory. Detailed radiation calculations remain State/National Unit 6; no numerical quantum calculation is assigned in this unit.